	GLOBAL ECONOMICS **Grade: 11th - Learning Evidence C5, C6 and C7** **Continuous distributions and sample data** **Teacher: Nicolás López Cuéllar**	**1st TERM** **2026–2027**

Learning objective: Use continuous probability models and sample data to describe typical values, variability, and interval probabilities, and interpret differences between sample estimates and theoretical values.	**Assessment criteria:** C5: Calculates and interprets the mean, variance and standard deviation of continuous distributions using standard formulas and functions. C6: Estimates interval probabilities using relative frequencies from generated samples. and compares them with theoretical probabilities. C7: Calculates sample means, variances, and standard deviations for different sample sizes and data representations.

Criteria assessment

This exam assesses C5, C6 and C7 separately, with 24 points per criterion (72 points in total). Each task shows its point allocation. Partial credit may be awarded when the correct procedure is clearly shown but arithmetic errors are made. Answers without the required procedure will not receive credit. Failure to write your name on the task sheet or submitting it after the deadline will result in a one-point deduction from each criterion. For this exam, a minimum of 15 out of 24 points is required to pass each criterion. Use unrounded intermediate values; round final statistics to three decimal places and percentage errors to two decimal places where necessary.

1. Two packing stations [C5-2]

A gift company uses two packing stations. At station A, an automated machine finishes a gift box in anywhere from 4 to 16 minutes. Within these limits, any two time windows of the same length are equally likely to contain the finishing time. Let A be its packing time in minutes.

At station B, staff pack custom gift boxes, also taking between 4 and 16 minutes. To describe many orders, the manager draws a smooth outline of how often different packing times occur. The outline starts at zero at 4 minutes, rises along a straight edge to its single highest point at 7 minutes, then falls along a straight edge to zero at 16 minutes. No packing times occur outside these limits. Let B be this packing time in minutes.

(a) Identify the distribution for A from the clues and calculate its mean, variance, and standard deviation using the corresponding formulas. **[1]**

(b) Identify the distribution for B from the clues and calculate its mean, variance, and standard deviation using the corresponding formulas. **[1]**

2. Two custom-order workshops [C5-2]

Two workshops make custom wooden signs. Each order takes between 2 and 11 hours to complete. The owners use smooth outlines to describe how often different completion times occur over many orders; times outside these limits never occur.

At workshop I, longer jobs are more common: the outline is a single straight edge rising from zero at 2 hours to its highest point at 11 hours. At workshop D, shorter jobs are more common: the outline is a single straight edge falling from its highest point at 2 hours to zero at 11 hours. Let I and D be the respective completion times in hours.

(a) Identify each distribution from the clues and calculate its mean, variance, and standard deviation using the corresponding formulas. **[2]**

3. Arrivals and daily revenue [C5-2]

A bicycle repair shop receives online service requests throughout its opening hours, averaging 8 requests per hour. Requests arrive independently, one at a time, and the arrival pace stays the same throughout the day. In any very short time window, the chance of a request is proportional to the window's length, while two or more requests in that window are negligible. How long the shop has already waited does not affect the chance of a request during the next minute. Let T be the time between consecutive requests in minutes.

On comparable working days, the shop's total revenue varies because of many small, independent influences, with none dominating. A smooth outline of daily revenues has a symmetric bell shape, highest at \$2400 and tapering on both sides. The two places where its sides change from becoming steeper to becoming flatter are \$2250 and \$2550. Treat this bell-shaped pattern as exact for the calculations. Let R be daily revenue measured in hundreds of dollars.

(a) Identify the distribution for T from the clues and use its formulas to find the mean, variance, and standard deviation in minutes. **[1]**

(b) Identify the revenue distribution from the clues. Use its formulas to find the mean, variance, and standard deviation in dollars. Interpret the mean. **[1]**

4. Water consumption, raw data [C6-1]

A simulated sample of daily household water consumption (hundreds of liters) is

2.1 2.5 2.9 3.0 3.2 3.6 3.9 4.0 4.3 4.7

2.4 2.8 3.1 3.4 3.7 4.0 4.1 4.5 4.8 5.2

The theoretical probability is $P(3 \le X \le 4) = 0.50$.

(a) Find the empirical probability of the stated interval from the given sample. **[0.5]**

(b) Calculate the absolute and percentage errors relative to the given theoretical probability. **[0.5]**

5. Service times, frequency data [C6-1]

Recorded service times are summarized below.

x_i (min)	3	6	9	12	15	18
f_i	6	18	30	24	15	7

The theoretical probability is $P(6 \le X < 15) = 0.75$.

(a) Find the empirical probability of the stated interval from the given sample. **[0.5]**

(b) Calculate the absolute and percentage errors relative to the given theoretical probability. **[0.5]**

6. Download times, grouped data [C6-2]

A sample of download times is summarized by the following frequencies. All times are in seconds; classes are left-inclusive and right-exclusive.

Class	$[0, 5)$	$[5, 10)$	$[10, 15)$	$[15, 20)$
f_i	4	8	6	2

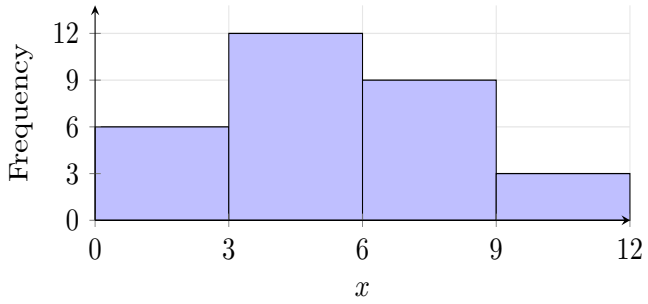
The model gives $P(5 \le X < 15) = 0.80$.

(a) Find the empirical probability of the stated interval from the given sample. **[1]**

(b) Calculate the absolute and percentage errors relative to the given theoretical probability. **[1]**

7. Ticket queues, histogram data [C6-2]

The histogram is the only record of a sample of queue times. Bar heights are frequencies; times are in minutes, with left-inclusive, right-exclusive classes.



The theoretical probability is $P(3 \le X < 9) = 0.75$.

(a) Find the empirical probability of the stated interval from the given sample. **[1]**

(b) Calculate the absolute and percentage errors relative to the given theoretical probability. **[1]**

Use the sample-variance denominator $n - 1$ throughout these four problems. For each statistic, calculate absolute error $E = |\text{sample value} - \text{theoretical value}|$ and percentage error $100E/|\text{theoretical value}|$. Use unrounded values in all error calculations. For grouped data and histograms, use class midpoints to estimate the statistics.

8. Sensor readings, raw data [C7-1]

Five sensor readings in degrees Celsius are $9, 11, 12, 13, 15$. The theoretical model has $\mu = 12$ degrees Celsius and $\sigma^2 = 4$ square degrees Celsius.

(a) Calculate the sample mean, sample variance, and sample standard deviation from the given data. **[0.5]**

(b) Calculate the absolute and percentage errors of the mean, variance, and standard deviation relative to the theoretical values. **[0.5]**

9. Items sold, frequency data [C7-1]

The numbers of items sold in six short sales periods are summarized below. The theoretical model has $\mu = 5$ items and $\sigma^2 = 5$ items^2.

x_i	2	5	8
f_i	1	3	2

(a) Calculate the sample mean, sample variance, and sample standard deviation from the given data. **[0.5]**

(b) Calculate the absolute and percentage errors of the mean, variance, and standard deviation relative to the theoretical values. **[0.5]**

10. Cycling times, grouped data [C7-2]

Cycling times in minutes are grouped as follows, using left-inclusive, right-exclusive classes. The theoretical mean is 18 minutes and variance is 20 minutes^2.

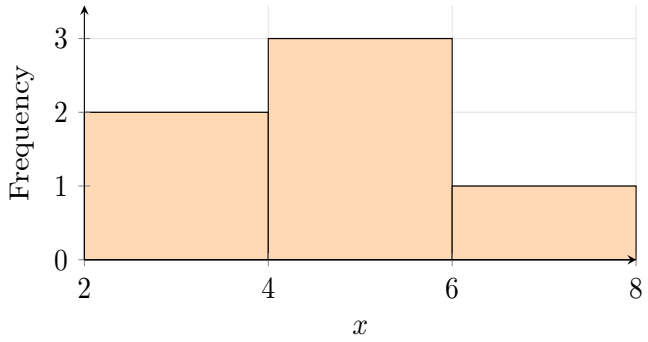
Time (min)	$[10, 14)$	$[14, 18)$	$[18, 22)$
f_i	1	2	3

(a) Calculate the sample mean, sample variance, and sample standard deviation from the given data. **[1]**

(b) Calculate the absolute and percentage errors of the mean, variance, and standard deviation relative to the theoretical values. **[1]**

11. Crate masses, histogram data [C7-2]

A histogram gives a sample of crate masses in kilograms. Bar heights are frequencies; classes are left-inclusive and right-exclusive. The model has $\mu = 5$ kg and $\sigma^2 = 2$ kg^2.



(a) Calculate the sample mean, sample variance, and sample standard deviation from the given data. **[1]**

(b) Calculate the absolute and percentage errors of the mean, variance, and standard deviation relative to the theoretical values. **[1]**